

30 Kubernetes Production Issues

Every DevOps / SRE Engineer Should Know

Causes & Fixes from Real Production Incidents



CrashLoopBackOff



OOMKilled



DNS Failures



Node NotReady



ImagePullBackOff

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POD FAILURES

Common Pod Failures

- **CrashLoopBackOff**
Cause: App crash or bad configuration
Fix: Check logs and environment variables
- **ImagePullBackOff**
Cause: Wrong image tag or registry auth issue
Fix: Verify image name and credentials
- **Pod Pending**
Cause: Insufficient resources or scheduling rules
Fix: Increase cluster capacity

RESOURCE ISSUES

Resource Related Problems

- **OOMKilled**
Cause: Memory limit exceeded
Fix: Increase memory or optimize application
- **CPU Throttling**
Cause: CPU limits too low
Fix: Adjust CPU limits and monitor usage
- **Node Disk Pressure**
Cause: Logs or images filling node disk
Fix: Clean logs and unused images

NETWORKING ISSUES

Networking Problems

- **DNS Resolution Failure**
Cause: CoreDNS crash or network policies
Fix: Restart CoreDNS and check policies
- **Service Discovery Failures**
Cause: Misconfigured services or ingress
Fix: Verify endpoints and DNS
- **Network Partition**
Cause: Node connectivity loss
Fix: Check routing and firewall rules

STORAGE ISSUES

Storage Related Issues:

- **PersistentVolume Pending**
Cause: Storage class mismatch
Fix: Verify storage class and PV availability
- **Disk Space Exhaustion**
Cause: Large volumes or logs
Fix: Expand storage or clean data
- **Quota Exhaustion**
Cause: Storage limits reached
Fix: Increase quotas

DATABASE PROBLEMS

Database Performance Issues:

- **Database Latency**
Cause: Slow queries
Fix: Optimize queries and indexing
- **Connection Leaks**
Cause: Too many open connections
Fix: Enable connection pooling
- **Indexing Issues**
Cause: Missing indexes
Fix: Add indexes and optimize queries

DEPLOYMENT FAILURES

Deployment Issues:

- **Canary Deployment Failure**
Cause: Traffic shifting misconfiguration
Fix: Validate rollout strategy
- **CI/CD Pipeline Failure**
Cause: Broken build artifacts
Fix: Add rollback strategy
- **Configuration Drift**
Cause: Manual changes in production
Fix: Use GitOps tools like ArgoCD

SECURITY & SECRETS

Security Related Incidents:

- **Secrets Mismanagement**
Cause: Hardcoded or expired secrets
Fix: Use Vault or Secrets Manager
- **SSL Certificate Expiry**
Cause: Certificates not renewed
Fix: Automate with Cert-Manager
- **Security Incidents**
Cause: Exposed IAM roles or ports
Fix: Follow least privilege

SCALING ISSUES

Scaling Problems:

- **Node Autoscaling Failure**
Cause: Cloud quota limits
Fix: Increase quotas
- **Insufficient IP Addresses**
Cause: CIDR exhaustion
Fix: Expand subnet range
- **Rate Limiting from APIs**
Cause: Too many requests
Fix: Add retry and caching

MONITORING & RELIABILITY

Observability Issues:

- **Alert Fatigue**
Cause: Too many alerts
Fix: Tune alert thresholds
- **Log Flooding**
Cause: Excess debug logs
Fix: Reduce log levels
- **Time Sync Issues**
Cause: NTP drift
Fix: Sync clocks across cluster